

Embedded Real-Time Programming

Candidate number: 8503

13/05/2026

Contents

1	Introduction	2
2	Methods	2
2.1	Hardware Setup	2
2.2	Software Setup	2
2.3	Model Training	3
3	Results	3
4	Discussion	5
5	Conclusion	5

AI Usage Statement: ChatGPT (GPT-5.5) was used for LaTeX assistance, improvement of academic language, grammar checking, shortening and restructuring the report to fit the page limit, and as support during development of the application code. All generated content and code suggestions were reviewed, fine-tuned, and validated by the author.

1 Introduction

The aim of the project was to design an embedded application capable of detecting plastic bottles in real time using the nRF54L15 DK. The project combined image acquisition, display output, and machine learning inference on the same embedded platform.

The application was developed using Zephyr RTOS together with Edge Impulse. An OV7670 camera module was used to capture images while a TFT display showed the camera feed and detection results.

Similar work has been presented in the BED project, where real-time object detection was implemented on the MAX78000 edge AI platform using a compact neural network model and on-device inference [3]. The BED project also combined camera input, display output, and embedded machine learning in a single system.

The main difference in this project is the use of the nRF54L15 DK together with the OV7670 camera and Edge Impulse FOMO architecture. Unlike BED, which focused on a custom compact DNN model, this project focused more on integrating Edge Impulse tools and lightweight embedded inference on Zephyr RTOS.

2 Methods

2.1 Hardware Setup

The main hardware platform used in the project was the Nordic Semiconductor nRF54L15 DK [1]. An OV7670 camera module with AL422B FIFO buffer was connected to the board for image capture. Images were captured in QQVGA resolution using RGB565 format.

A 1.8-inch TFT display with an ST7735R controller was connected through SPI and used to display the live camera feed and object detection results.

The system was powered and programmed through USB connection to a Windows 11 computer. Serial communication at 115200 baud was used for debugging and monitoring.

2.2 Software Setup

The firmware was developed in Visual Studio Code using the nRF Connect SDK and Zephyr RTOS. The application was written in C/C++ and compiled using the `west` build system.

The project combined provided OV7670 camera and TFT display example projects from the course repository into a single application [4]. The display used the Zephyr ST7735R driver together with custom drawing functions for image rendering and bounding boxes. The complete project source code, including the integrated camera, display, and AI inference application, is available on GitHub: <https://github.com/gzupan00/nrf54l15-edge-ai-project>

Edge Impulse was used for dataset preparation, model training, and model export [2]. The final AI model was integrated into the firmware as a Zephyr-compatible library.

2.3 Model Training

The object detection model was trained using Edge Impulse with the FOMO architecture. The dataset consisted of 616 manually collected images containing different plastic bottles under different lighting conditions and backgrounds.

All images were labeled manually using bounding boxes with a single class label called **plastic bottle**. The dataset was divided into approximately 80% training data and 20% testing data.

Images were resized and converted to grayscale automatically inside Edge Impulse. The final input resolution was reduced to 50×50 pixels to fit within the memory limitations of the embedded hardware.

The model was trained for 30 epochs using default Edge Impulse augmentation settings with a learning rate of 0.001. During training, both training loss and validation loss gradually decreased and later stabilized.

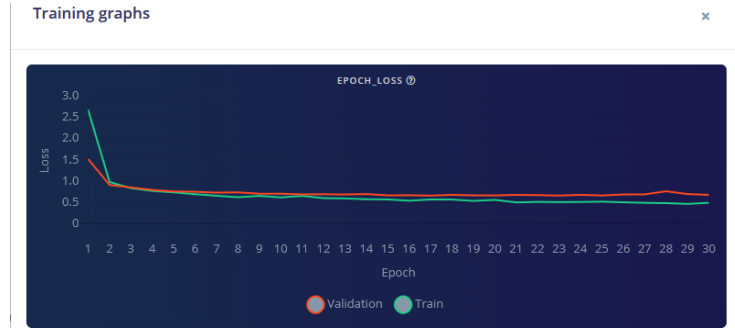


Figure 1: Training and validation loss during model training.

3 Results

The final FOMO object detection model achieved an accuracy of 52.42%. This means that 52.42% of detections produced confidence values above 80% during evaluation in Edge Impulse.

The evaluation metrics included a precision value of 0.76 and recall value of 0.54 for non-background object detection.

Results		Model version: Quantized (int8)
% ACCURACY 52.42%		
Metrics for Object detection		
METRIC	VALUE	
Precision (non-background)	0.76	
Recall (non-background)	0.54	
F1 Score (non-background)	0.63	

Figure 2: Final object detection accuracy and evaluation metrics.

The embedded system successfully displayed bounding boxes around detected bottles on the TFT display using live camera input from the OV7670 module.



Figure 3: Examples of object detection results displayed on the TFT screen.

Serial monitor output confirmed successful inference execution and displayed detection confidence values together with bounding box coordinates.

```

Button pressed - running inference...
Inference done err=0 boxes=1
BB label=plastic bottle value=92% x=21 y=28 w=7 h=7
Bottle detected: 92%
Button pressed - running inference...
Inference done err=0 boxes=1
BB label=plastic bottle value=96% x=21 y=28 w=7 h=7
Bottle detected: 96%
Button pressed - running inference...
Inference done err=0 boxes=0

```

Figure 4: Serial monitor output showing successful detections.

4 Discussion

The model was generally able to detect plastic bottles successfully, especially under stable lighting conditions. However, missed detections and false positives occurred in more complex environments.

The main challenge in the data collection phase was the poor image quality from the camera module, which resulted in about 30% of the images being unusable.

In the development phase the main concern was the limited RAM available on the embedded hardware. That was why the image resolution, on which model was trained, had to be lowered, and that resulted in a performance drop when it came to inference.

In the deployment phase the limitation was the image quality from the camera module, which affected detection accuracy under poor lighting conditions.

Despite these limitations, the project successfully integrated camera communication, display output, and machine learning inference into a functional embedded application.

5 Conclusion

This project presented the development of a real-time embedded object detection system using the nRF54L15 DK and Edge Impulse FOMO model.

The final system successfully displayed live video output, and performed object detection directly on the embedded hardware. The project demonstrated that embedded machine learning inference can be implemented on resource-constrained systems using lightweight AI models.

References

- [1] Nordic Semiconductor, *nRF54L15 Development Kit Documentation*. Available: <https://www.nordicsemi.com/Products/Development-hardware/nRF54L15-DK>
- [2] Edge Impulse, *FOMO Object Detection Documentation*. Available: <https://docs.edgeimpulse.com/>
- [3] G. Wang, Z. P. Bhat, Z. Jiang et al., “BED: A Real-Time Object Detection System for Edge Devices,” in *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM '22)*, 2022.
- [4] S. Bos, *USN-CS4140-EmbeddedAI*, GitHub repository. Available: <https://github.com/aiunderstand/USN-CS4140-EmbeddedAI>
- [5] Trond Elias Eide Roaas and Nicolas Theodor Fafalios Olsen, assistance with OV7670 camera debugging and communication fixes.